Mobility, Economic Conditions, and Crime During COVID-19: Evidence from Boston

Course: ECO225
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Project one

1.1 Introduction

Research on crime has long examined how economic conditions shape offending, with many studies focusing on unemployment, labor market stress, and opportunity cost. Classic economic models emphasize that weaker labor market opportunities may reduce the opportunity cost of offending and increase incentives for crime (Becker, 1968). Raphael and Winter-Ebmer (2001) find evidence that unemployment increases crime, while Gould, Weinberg, and Mustard (2002) show that weaker labor market opportunities are associated with higher criminal activity. This project starts from that framework, asking not just whether reported crime in Boston rose or fell during COVID-19, but how its composition changed over time as both economic conditions and daily movement patterns were disrupted.

That shift matters because COVID-19 was not only a labor market shock. It also changed how people moved through the city, how often they used public transit, and how often potential offenders, victims, and guardians came into contact. Research in environmental criminology and routine activity theory emphasizes that crime opportunities depend on the convergence of these elements in space and time (Cohen and Felson, 1979; Felson and Boba, 2010). Newton's overview of the urban mobility literature highlights that crime patterns are connected to the movement of people through urban space, making mobility an important explanatory factor.

This perspective is especially useful for studying crime during the pandemic. Mohler et al. argue that social distancing could change both the volume and distribution of crime by increasing guardianship in some settings while reducing public interaction in others. Hodgkinson and Andresen similarly find that the pandemic affected crime types unevenly, supporting the idea that shifts in activity patterns and opportunity structures matter for understanding crime during extraordinary events. Hoeboer et al. conclude that many crime categories fell during COVID restrictions, though the effects differed across offense types and often changed once restrictions eased. Ashby (2020) documents early evidence that crime declined during lockdowns in several U.S. cities, while Abrams (2021) provides one of the first empirical analyses linking COVID restrictions to crime changes.

More recent evidence strengthens this framing. Frith, Bowers, and Johnson show that changes in household occupancy during COVID restrictions help explain burglary patterns, while Trajtenberg et al. find across 45 cities that stricter lockdowns were associated with larger declines in robbery, burglary, and vehicle theft, but not all crime types. Studies of major U.S. cities also emphasize that mobility reductions and changes in routine activity were central mechanisms shaping these outcomes (Campedelli, Aziani, and Favarin, 2021). Together, these studies suggest that pandemic-era crime should be understood through both economic stress and changing patterns of mobility and routine activity.

This project applies that framework to Boston using monthly Boston Police Department incident data from 2018 to 2022. It examines how reported crime and its composition changed before, during, and after the main COVID disruption. The analysis combines monthly crime counts with unemployment, labor force, a COVID indicator, a time trend, and a Google mobility measure based on transit station activity. The goal is descriptive rather than strictly causal: to test whether changes in local economic conditions and mobility are systematically associated with changes in reported crime over time.

1.2 Data Loading and Cleaning

The analysis combines three Boston datasets: police-reported crime, labor market conditions, and mobility. The crime data were collected from separate Boston Police Department files and merged into a single dataset after standardizing column names and cleaning date fields. Invalid observations were removed, and the incident date was used to create year and month variables so the data could be aggregated to the monthly level. In addition to total monthly crime, broad crime categories were also constructed in order to track changes in crime composition over time.

Labor market data were merged in using monthly Boston-level measures from FRED, including the unemployment rate and labor force. The Google Community Mobility dataset was cleaned from its broader global file and filtered to the Boston area, then aggregated to monthly transit mobility values. After merging the three sources, a COVID-period indicator and a linear time trend were created. The result is a monthly Boston panel from 2018 to 2022 that supports both the descriptive analysis and the regression section.

Imports:

In [176…

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import statsmodels.api as sm
import glob
import os
```

Loading and combing yearly Boston crime files:

In [177…

```
crime_files = [
    "2018 Boston crime.csv",
    "2019 Boston crime.csv",
    "2020 Boston crime.csv",
    "2021 Boston crime.csv",
    "2022 Boston crime.csv"
]

crime_dfs = []
for file in crime_files:
    df = pd.read_csv(file, low_memory=False)
    df.columns = df.columns.str.lower().str.strip()
    crime_dfs.append(df)

crime_raw = pd.concat(crime_dfs, ignore_index=True)
crime_raw = crime_raw.loc[:, ~crime_raw.columns.duplicated()]
```

Cleaning dates and creating time variables:

In [178…

```
crime_raw["occurred_on_date"] = pd.to_datetime(crime_raw["occurred_on_date"], errors="coerce")
crime_raw = crime_raw.dropna(subset=["occurred_on_date"])
```

```
crime_raw["year"] = crime_raw["occurred_on_date"].dt.year
crime_raw["month"] = crime_raw["occurred_on_date"].dt.month
```

Creating monthly total crimes:

In [179…

```
monthly_crime = (
    crime_raw.groupby(["year", "month"])
    .size()
    .reset_index(name="crime_count")
)
```

Creating crime composition categories:

In [260…

```
df = crime_raw.copy()

df["offense_description"] = df["offense_description"].astype(str).str.strip().str.lower()
df["offense_description"] = (
    df["offense_description"]
    .replace({"nan": "other", "": "other"})
    .fillna("other")
)

def crime_group(desc: str) -> str:
    if "assault" in desc or "harassment" in desc or "threat" in desc:
        return "violence_assault"
    elif "larceny" in desc or "vandalism" in desc or "burglary" in desc \
        or "property - lost" in desc or "fraud" in desc:
        return "property_related"
    elif "motor vehicle" in desc or "m/v" in desc or "leaving scene" in desc \
        or "towed" in desc or "auto theft" in desc:
        return "vehicle_related"
    elif "investigate" in desc:
        return "investigation"
    elif "sick" in desc or "medical" in desc:
        return "medical_assist"
    elif "drug" in desc or "narcotic" in desc:
        return "drug_related"
    else:
        return "other"

df["crime_group"] = df["offense_description"].apply(crime_group)

monthly_crime_group = (
    df.groupby(["year", "month", "crime_group"])
      .size()
      .reset_index(name="crime_count")
)

monthly_crime_wide = (
    monthly_crime_group
    .pivot_table(
        index=["year", "month"],
        columns="crime_group",
        values="crime_count",
        aggfunc="sum",
        fill_value=0
    )
    .reset_index()
)
```

```
monthly_crime_wide["total_crime"] = \
monthly_crime_wide.drop(columns=["year", "month"]).sum(axis=1)
```

Cleaning labor market data:

In [181…

```
def clean_fred_series(df, value_name):
    df["observation_date"] = pd.to_datetime(df["observation_date"])
    df["year"] = df["observation_date"].dt.year
    df["month"] = df["observation_date"].dt.month
    df = df.rename(columns={df.columns[1]: value_name})
    return df[["year", "month", value_name]]

df_rate = pd.read_csv("Boston unemployment rate 1990-2025.csv")
df_force = pd.read_csv("Boston labor force 1990-2025.csv")
df_unemp = pd.read_csv("Boston unemployment 1990-2025.csv")

econ_rate = clean_fred_series(df_rate, "unemployment_rate")
econ_force = clean_fred_series(df_force, "labor_force")
econ_unemp = clean_fred_series(df_unemp, "unemployed_count")

boston_econ = econ_rate.merge(econ_force, on=["year", "month"]).merge(
    econ_unemp, on=["year", "month"]
)

boston_econ = boston_econ.query("2018 <= year <= 2022")
```

Cleaning Boston mobility global mobility data:

In [182…

```
mobility = pd.read_csv("Global_Mobility_Report.csv", low_memory=False)
mobility.columns = mobility.columns.str.lower().str.strip()

boston_mobility = mobility[
    (mobility["country_region"] == "United States") &
    (mobility["sub_region_1"] == "Massachusetts") &
    (mobility["sub_region_2"] == "Suffolk County")
].copy()

boston_mobility["date"] = pd.to_datetime(boston_mobility["date"])
boston_mobility["year"] = boston_mobility["date"].dt.year
boston_mobility["month"] = boston_mobility["date"].dt.month

boston_mobility_monthly = (
    boston_mobility.groupby(["year", "month"])[
        "transit_stations_percent_change_from_baseline"
    ]
    .mean()
    .reset_index()
)
```

Merging final analysis dataset + Covid dummy

In [183…

```
merged_df = (
    monthly_crime_wide
    .merge(boston_econ, on=["year", "month"], how="inner")
    .merge(boston_mobility_monthly, on=["year", "month"], how="left")
)

merged_df["date"] = pd.to_datetime(
    dict(year=merged_df["year"], month=merged_df["month"], day=1)
```

```
)
merged_df = merged_df.sort_values("date").reset_index(drop=True)

merged_df["covid"] = (merged_df["date"] >= "2020-03-01").astype(int)
merged_df["trend"] = np.arange(1, len(merged_df) + 1)
```

1.3 Summary Statistics Tables

In [184…

```
import pandas as pd
from IPython.display import HTML, display

summary_vars = [
    "total_crime",
    "unemployment_rate",
    "covid",
    "trend",
    "transit_stations_percent_change_from_baseline",
    "labor_force"
]

var_labels = {
    "total_crime": "Monthly reported crime",
    "unemployment_rate": "Unemployment rate (%)",
    "covid": "COVID period dummy",
    "trend": "Time trend",
    "transit_stations_percent_change_from_baseline": "Transit mobility change (%)",
    "labor_force": "Labor force"
}

summary_table = (
    merged_df[summary_vars]
    .agg(["count", "mean", "std", "min", "max"])
    .T
    .reset_index()
    .rename(columns={
        "index": "Variable",
        "count": "N",
        "mean": "Mean",
        "std": "Std. Dev.",
        "min": "Min",
        "max": "Max"
    })
    .round(2)
)

summary_table["Variable"] = summary_table["Variable"].map(var_labels)

display(HTML("<h4>Table 1. Summary Statistics</h4>" + summary_table.to_html(index=False)))
```

Table 1. Summary Statistics

Variable	N	Mean	Std. Dev.	Min	Max
Monthly reported crime	60.0	6708.98	1098.57	4387.00	9072.0
Unemployment rate (%)	60.0	4.62	3.10	2.60	16.5
COVID period dummy	60.0	0.57	0.50	0.00	1.0
Time trend	60.0	30.50	17.46	1.00	60.0
Transit mobility change (%)	11.0	-56.81	21.49	-80.03	0.4
Labor force	60.0	2803432.10	46986.74	2595759.00	2883061.0

Table 1 shows substantial variation in the main variables used in the analysis. Monthly reported crime averages about 6,709 incidents, with values ranging from 4,387 to 9,072, suggesting that crime levels changed meaningfully over the sample period. The unemployment rate varies widely, from 2.6% to 16.5%, capturing the major labor market disruption associated with COVID-19. Labor force levels are more stable but still reflect meaningful shifts during the pandemic years. Transit mobility, available only for 2020, shows a large average decline of about 57% from baseline levels, which is consistent with the broader disruption in urban movement during the lockdown period.

In [185…

```
subgroup_df = merged_df.copy()
subgroup_df["date"] = pd.to_datetime(subgroup_df["date"])

def period_group(date):
    if date < pd.Timestamp("2020-03-01"):
        return "Pre-COVID"
    elif date <= pd.Timestamp("2020-12-31"):
        return "COVID Shock"
    else:
        return "Post-COVID"

subgroup_df["period_group"] = subgroup_df["date"].apply(period_group)

subgroup_table = (
    subgroup_df.groupby("period_group")[[
        "total_crime",
        "unemployment_rate",
        "labor_force",
        "transit_stations_percent_change_from_baseline"
    ]]
    .mean()
    .round(2)
    .reset_index()
)

# fill missing mobility values for cleaner display
subgroup_table["transit_stations_percent_change_from_baseline"] = (
    subgroup_table["transit_stations_percent_change_from_baseline"].fillna("")
)

# order rows
order = ["Pre-COVID", "COVID Shock", "Post-COVID"]
subgroup_table["period_group"] = pd.Categorical(
    subgroup_table["period_group"],
    categories=order,
```

```
    ordered=True
)

subgroup_table = subgroup_table.sort_values("period_group").reset_index(drop=True)

subgroup_table = subgroup_table.rename(columns={
    "transit_stations_percent_change_from_baseline": "Transit Mobility (%)"})

display(HTML("<h4>Table 1A. Means by Period</h4>" + subgroup_table.to_html(index=False)))
```

Table 1A. Means by Period

period_group	total_crime	unemployment_rate	labor_force	Transit Mobility (%)
Pre-COVID	7627.35	2.92	2839843.65	0.4
COVID Shock	5865.50	10.23	2770491.90	-62.53
Post-COVID	6065.54	4.12	2777711.33	

Note: Transit mobility data are only available for 11 months in 2022, so mobility averages are reported only for the months where the data exist.

Table 1A breaks the sample into three periods: Pre-COVID (before March 2020), COVID Shock (March to December 2020), and Post-COVID (2021 onwards). Average reported crime falls from about 7,627 incidents per month pre-COVID to about 5,866 during the shock period, even as the unemployment rate rises sharply from 2.92% to 10.23% and labor force participation declines. This pattern is the central puzzle the paper addresses: crime fell at exactly the time economic conditions worsened most severely. Reported crime remains below its pre-COVID level in the post-COVID period, suggesting only a partial recovery. Transit mobility data are only available during the COVID shock period, where they show an average decline of about 63% from baseline, reinforcing the idea that reduced urban movement coincided with falling crime.

1.4 Plots, Histograms, Figures

Figure 1 plots monthly reported crime in Boston from 2018 to 2022. Crime is relatively high before 2020, drops sharply at the onset of COVID-19, and then recovers only partially over the following two years. The pattern suggests a clear structural break around the pandemic rather than a small month-to-month fluctuation. The earlier peaks and troughs also hint at seasonal variation, but the decline in early 2020 is much larger than the normal seasonal movement, which is why COVID is treated as a distinct shock in the analysis.

In [186…

```
fig1_df = merged_df.copy()
fig1_df["date"] = pd.to_datetime(fig1_df["date"])
fig1_df = fig1_df.sort_values("date")

plt.figure(figsize=(11, 5.5))
plt.plot(fig1_df["date"], fig1_df["total_crime"], linewidth=2, color="firebrick")
plt.axvline(pd.Timestamp("2020-03-01"), linestyle="--", linewidth=1.5, color="red")
plt.text(pd.Timestamp("2020-04-01"), fig1_df["total_crime"].max() * 0.97, "COVID onset")

plt.title("Figure 1. Monthly Reported Crime in Boston, 2018–2022")
plt.xlabel("Date")
plt.ylabel("Reported incidents")
plt.tight_layout()
plt.show()
```

Figure 1. Monthly Reported Crime in Boston, 2018–2022

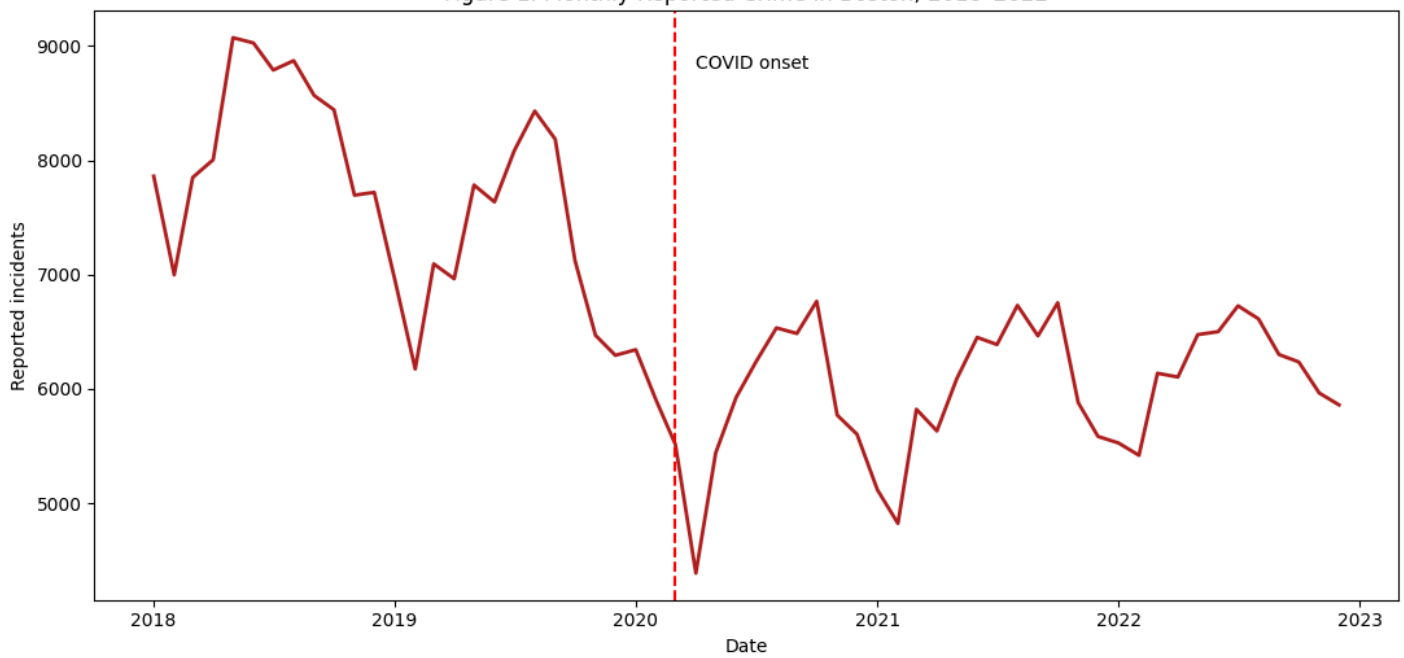


Figure 2 shows that Boston's labor force went through a sharp break at the start of the COVID-19 period. Around March 2020, labor force participation drops abruptly, indicating the immediate disruption the pandemic created in the local economy. Although the series recovers somewhat afterward, it does not return to its pre-shock peak. This figure matters because it shows that COVID affected economic conditions through more than just the unemployment rate. It also reduced overall labor market participation, which is why labor force is included separately in the analysis rather than treating unemployment as the only relevant economic variable.

In [187…

```
plt.figure(figsize=(10,5))

plt.plot(merged_df["date"], merged_df["labor_force"], label="Labor force")

plt.axvline(pd.to_datetime("2020-03-01"), linestyle="--", color="red")

plt.title("Figure 2. Boston Labor Force During the COVID Shock")
plt.xlabel("Date")
plt.ylabel("Labor force")
plt.legend()

plt.show()
```

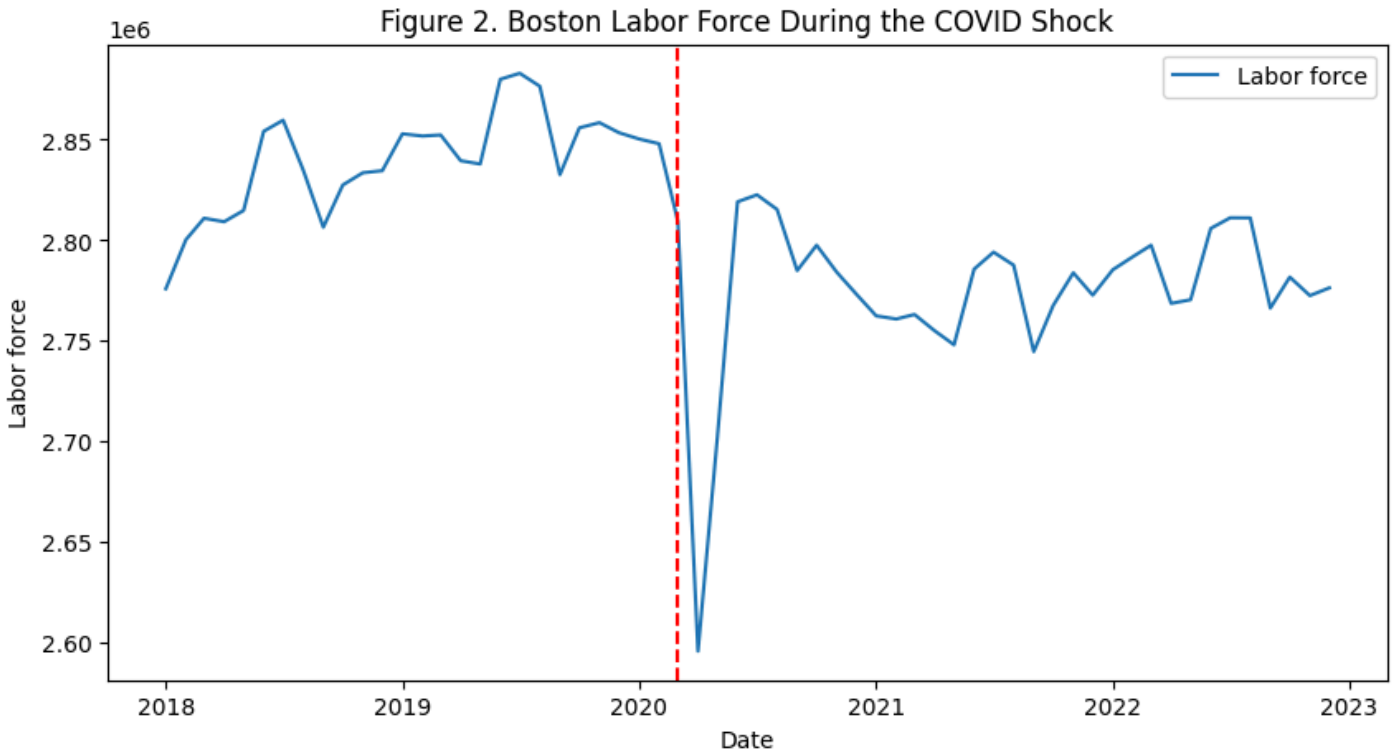


Figure 3 presents a scatter plot of total monthly crime against the unemployment rate. The relationship appears weakly negative, with higher unemployment rates associated with lower levels of reported crime. This pattern likely reflects reduced mobility and social interaction during periods of economic distress, rather than a direct causal effect of unemployment on crime. The spread of the points suggests that unemployment alone cannot explain variation in crime, which motivates the later multivariate analysis.

In [188…

```
plt.figure(figsize=(8,5))
plt.scatter(
    merged_df["unemployment_rate"],
    merged_df["total_crime"],
    alpha=0.7,
    color="firebrick"
)

plt.title("Figure 3. Total Crime vs Unemployment Rate")
plt.xlabel("Unemployment Rate (%)")
plt.ylabel("Total Monthly Crime")
plt.tight_layout()
plt.show()
```

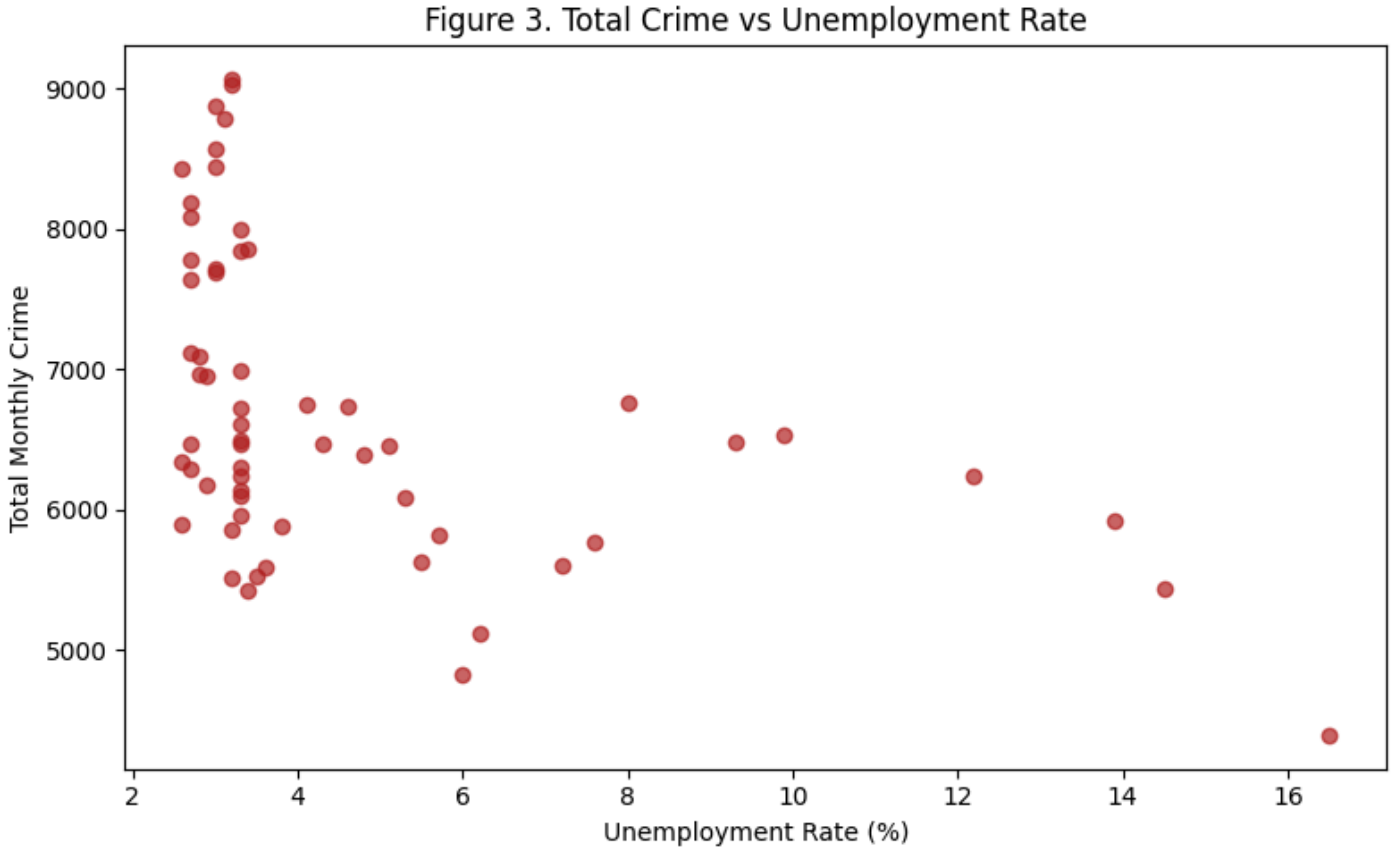
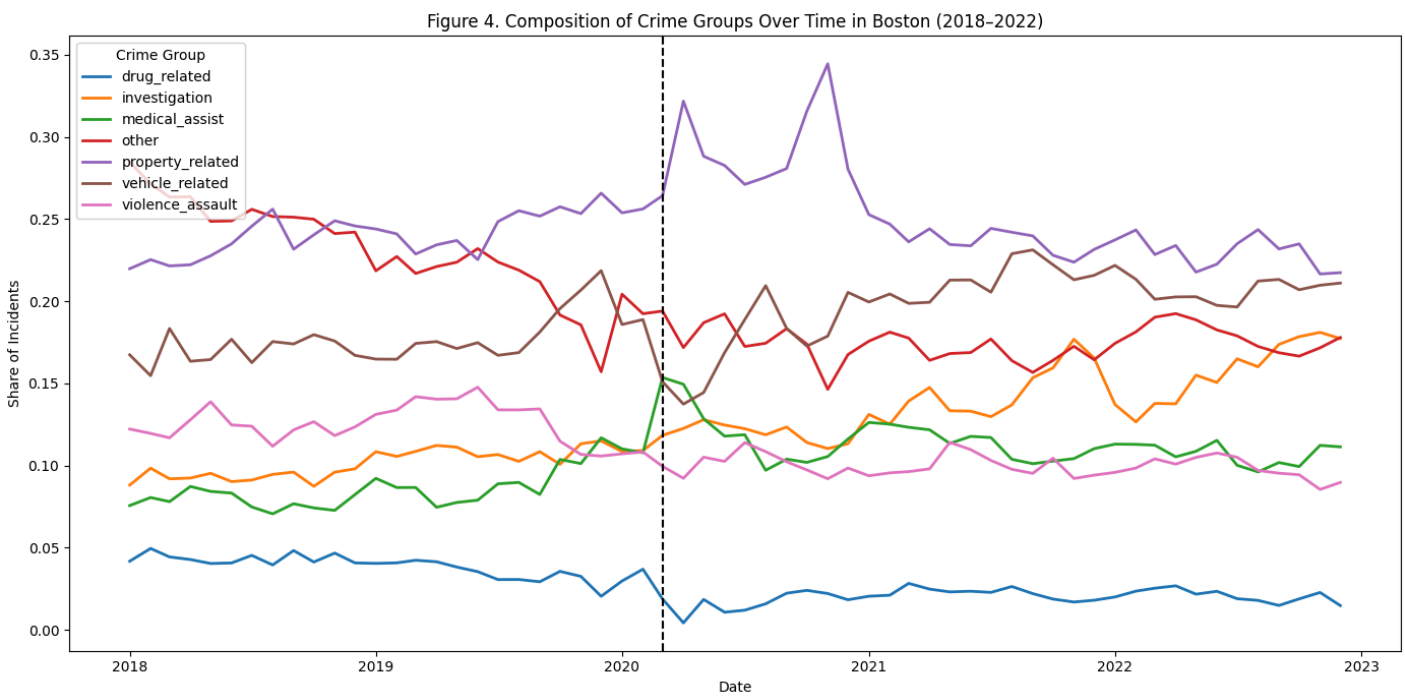


Figure 4 shows how the composition of reported crime changed across major categories over time. Unlike the total-crime plot, this figure focuses on shares rather than levels, making it easier to see whether some categories became more or less prominent during the pandemic period. A particularly noticeable break appears around March 2020. Property crime rises as a share of total incidents at that point, while vehicle-related crime drops sharply. This pattern suggests that the lockdown period did not affect all types of crime in the same way. Reduced commuting, lower traffic volume, and changes in daily routines likely altered the opportunity structure for different offenses. Overall, the figure supports the idea that COVID changed not only the level of reported crime, but also its composition.

In [189…

```
df["occurred_on_date"] = pd.to_datetime(df["occurred_on_date"], errors="coerce")
df = df.dropna(subset=["occurred_on_date"])

df["date"] = df["occurred_on_date"].dt.to_period("M").dt.to_timestamp()

cat_col = "crime_group"

m_counts = (
    df.groupby(["date", cat_col])
      .size()
      .reset_index(name="incidents")
)

m_counts["share"] = (
    m_counts["incidents"] /
    m_counts.groupby("date")["incidents"].transform("sum")
)

share_wide = (
    m_counts.pivot(index="date", columns=cat_col, values="share")
            .fillna(0)
            .sort_index()
)
```

```python
desired_order = [
    "Property Crime",
    "Assault / Violence",
    "Vehicle Related",
    "Investigation",
    "Medical / Assist",
    "Other"
]

cols = [c for c in desired_order if c in share_wide.columns]

if len(cols) > 0:
    share_wide = share_wide[cols]

plt.figure(figsize=(14, 7))

for col in share_wide.columns:
    plt.plot(share_wide.index, share_wide[col], label=col, linewidth=2)

plt.axvline(pd.Timestamp("2020-03-01"), color="black", linestyle="--", linewidth=1.5)
plt.title("Figure 4. Composition of Crime Groups Over Time in Boston (2018–2022)")
plt.xlabel("Date")
plt.ylabel("Share of Incidents")

if len(share_wide.columns) > 0:
    plt.legend(title="Crime Group", loc="upper left")

plt.tight_layout()
plt.show()
```

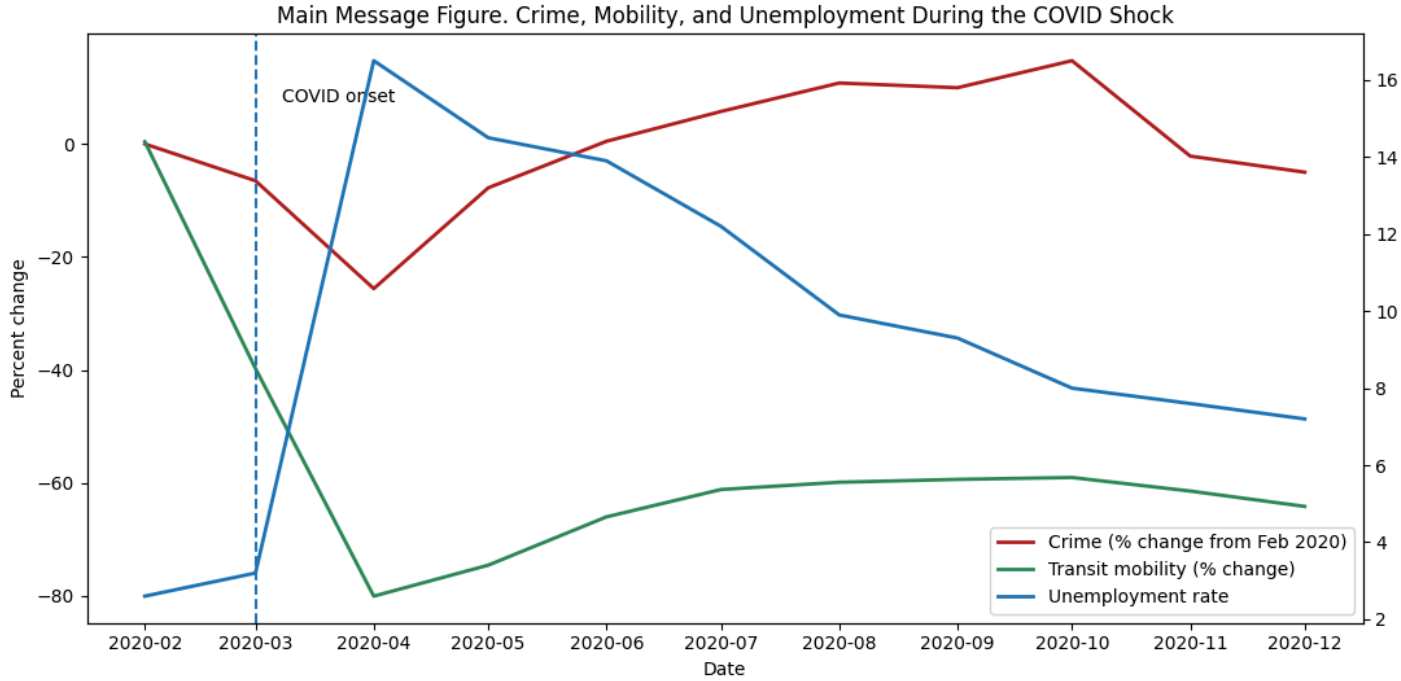


Project 2

2.1 The Message: Crime fell during the COVID-19 lockdown despite rising unemployment, suggesting mobility restrictions outweighed economic distress in the short run.

In [190…

```
fig_mob = merged_df.dropna(
    subset=["transit_stations_percent_change_from_baseline"]
).copy()

fig_mob["date"] = pd.to_datetime(fig_mob["date"])
fig_mob = fig_mob.sort_values("date").reset_index(drop=True)

baseline_crime = fig_mob.loc[
    fig_mob["date"] == pd.Timestamp("2020-02-01"),
    "total_crime"
].iloc[0]

fig_mob["crime_pct_change"] = (
    (fig_mob["total_crime"] - baseline_crime) / baseline_crime
) * 100
fig, ax1 = plt.subplots(figsize=(11,5.5))

ax1.plot(
    fig_mob["date"],
    fig_mob["crime_pct_change"],
    linewidth=2,
    label="Crime (% change from Feb 2020)",
    color="firebrick"
)

ax1.plot(
    fig_mob["date"],
    fig_mob["transit_stations_percent_change_from_baseline"],
    linewidth=2,
    label="Transit mobility (% change)",
    color="seagreen"
)

ax1.set_ylabel("Percent change")
ax1.set_xlabel("Date")

ax2 = ax1.twinx()

ax2.plot(
    fig_mob["date"],
    fig_mob["unemployment_rate"],
    linewidth=2,
    label="Unemployment rate"
)


ax1.axvline(pd.Timestamp("2020-03-01"), linestyle="--", linewidth=1.5)
ax1.text(
    pd.Timestamp("2020-03-08"),
    fig_mob["crime_pct_change"].max()*0.5,
    "COVID onset"
)

lines, labels = ax1.get_legend_handles_labels()
lines2, labels2 = ax2.get_legend_handles_labels()

ax1.legend(lines + lines2, labels + labels2)

plt.title("Main Message Figure. Crime, Mobility, and Unemployment During the COVID Shock")
plt.tight_layout()
plt.show()
```



The main message figure plots three series together over the 2020 mobility window: crime as a percent change from February 2020, transit mobility as a percent change from baseline, and the unemployment rate. I chose these three series together because they show the broader disruption caused by COVID-19 more clearly than any one variable alone. Around March 2020, reported crime falls sharply, transit mobility drops even more dramatically, and unemployment rises at the same time. This makes clear that the pandemic was not only a labor market shock, but also a major disruption to routine movement through the city. The three-series figure captures the central tension of the paper: crime declines even as economic distress worsens, suggesting that mobility restrictions may have mattered more than unemployment in the short run.

2.2 Maps and Interpretations

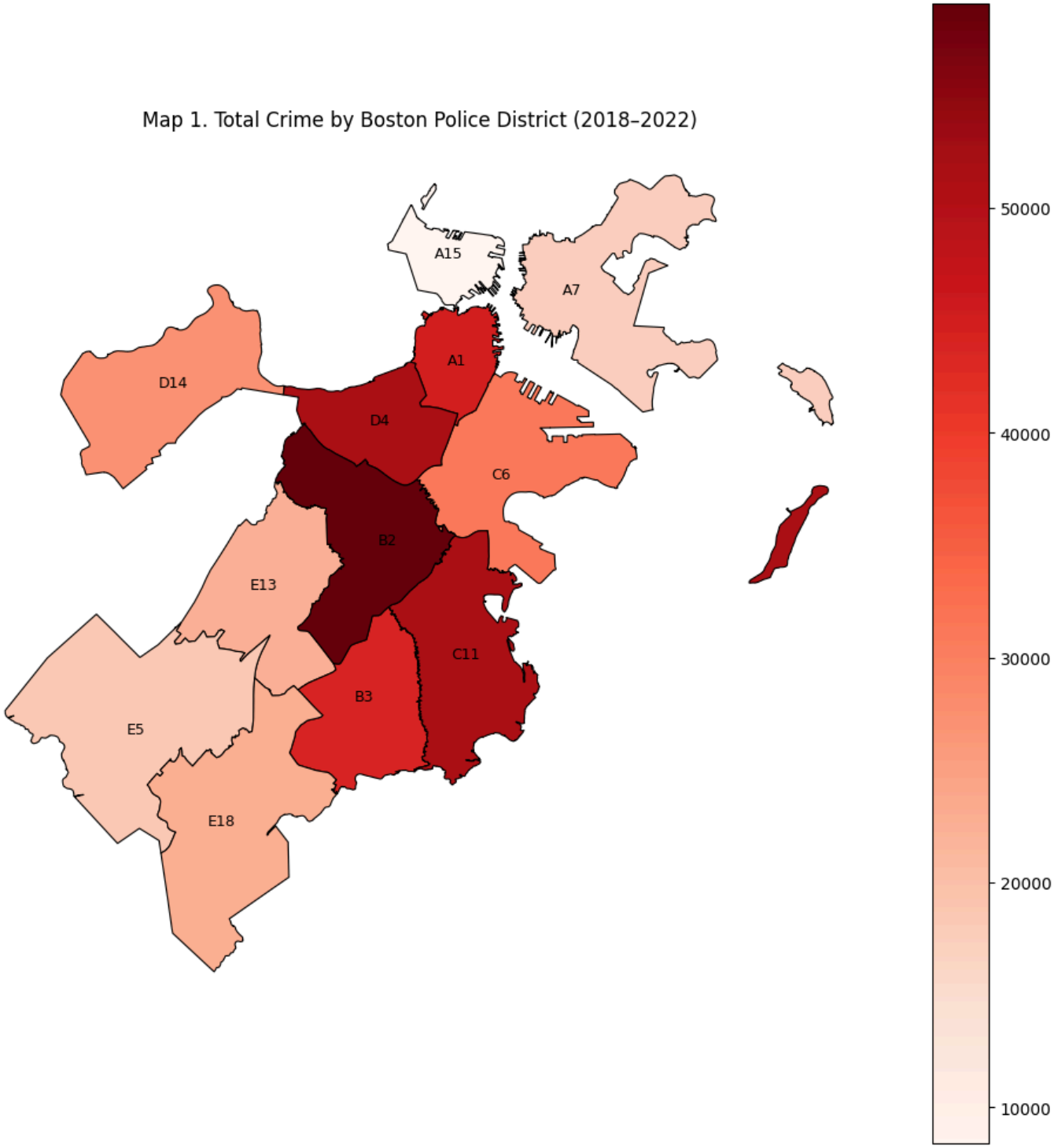
In [191…

```
import geopandas as gpd

district_map = gpd.read_file("boston_police_districts.geojson")

district_map["DISTRICT"] = district_map["DISTRICT"].astype(str).str.strip().str.upper()

crime_raw["district"] = crime_raw["district"].astype(str).str.strip().str.upper()

crime_clean = crime_raw[crime_raw["district"] != "EXTERNAL"].copy()

district_counts = (
    crime_clean.groupby("district")
    .size()
    .reset_index(name="crime_count")
)

map_data = district_map.merge(
    district_counts,
    left_on="DISTRICT",
    right_on="district",
    how="left"
)
```

In [192…

```
fig, ax = plt.subplots(figsize=(10,10))
```

```python
map_data.plot(
    column="crime_count",
    cmap="Reds",
    linewidth=0.8,
    edgecolor="black",
    legend=True,
    ax=ax
)

for idx, row in map_data.iterrows():
    point = row.geometry.representative_point()
    ax.annotate(
        text=row["DISTRICT"],
        xy=(point.x, point.y),
        ha="center",
        fontsize=9
    )


ax.set_title("Map 1. Total Crime by Boston Police District (2018–2022)")
ax.axis("off")

plt.tight_layout()
plt.show()
```



Map 1 shows the average level of reported crime across Boston police districts over the sample period. The spatial pattern is not uniform, with some districts consistently recording higher levels of crime than others (Central districts). This matters because it suggests that crime in Boston is geographically concentrated rather than spread evenly across the city. And this map provides a baseline spatial picture for the project and helps identify which districts remain most important in the later comparisons.

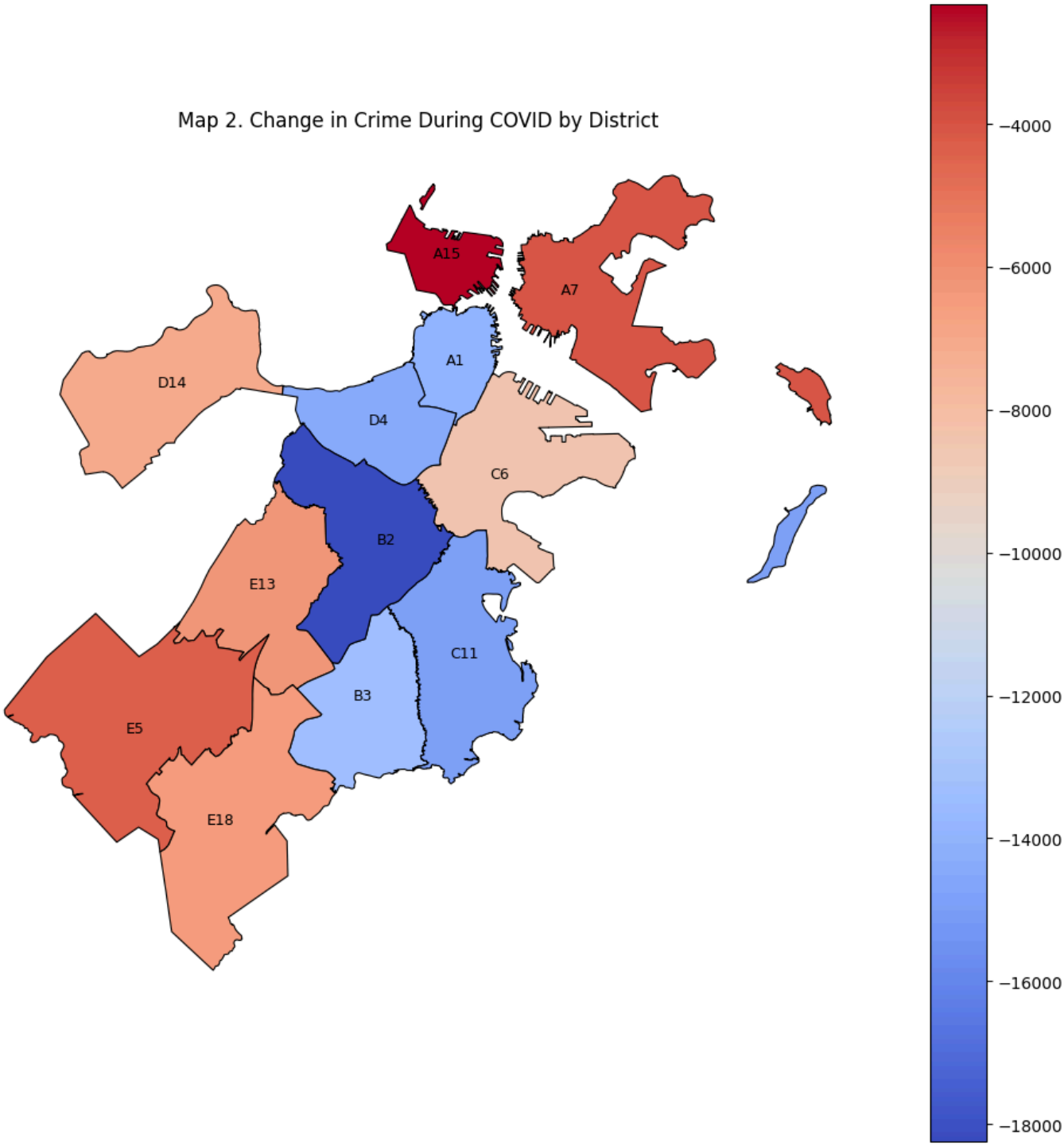
In [193…

```
crime_clean["period"] = "post"
crime_clean.loc[crime_clean["year"].isin([2018,2019]), "period"] = "pre"
crime_clean.loc[crime_clean["year"] == 2020, "period"] = "covid"
district_period = (
    crime_clean.groupby(["district","period"])
    .size()
    .reset_index(name="crime_count")
)
```

```
district_period = district_period.pivot(
    index="district",
    columns="period",
    values="crime_count"
).reset_index()
district_period["covid_change"] = (
    district_period["covid"] - district_period["pre"]
)
map_change = district_map.merge(
    district_period,
    left_on="DISTRICT",
    right_on="district",
    how="left"
)
```

In [194…

```
fig, ax = plt.subplots(figsize=(10,10))

map_change.plot(
    column="covid_change",
    cmap="coolwarm",
    linewidth=0.8,
    edgecolor="black",
    legend=True,
    ax=ax
)

for idx, row in map_change.iterrows():
    point = row.geometry.representative_point()
    ax.annotate(
        text=row["DISTRICT"],
        xy=(point.x, point.y),
        ha="center",
        fontsize=9
    )

ax.set_title("Map 2. Change in Crime During COVID by District")
ax.axis("off")

plt.tight_layout()
plt.show()
```



Map 2 shows how reported crime changed across police districts between the pre-COVID and COVID periods. The distribution of change is uneven, suggesting that the pandemic did not affect all parts of the city in the same way. Some districts experienced sharper declines than others (B2, C11, D4, A1, B3), which is consistent with the broader argument that the COVID shock altered routine activity, movement, and reporting patterns differently across space. This map complements the time-series figures by showing that the disruption was not only temporal but also geographic.

In [195…

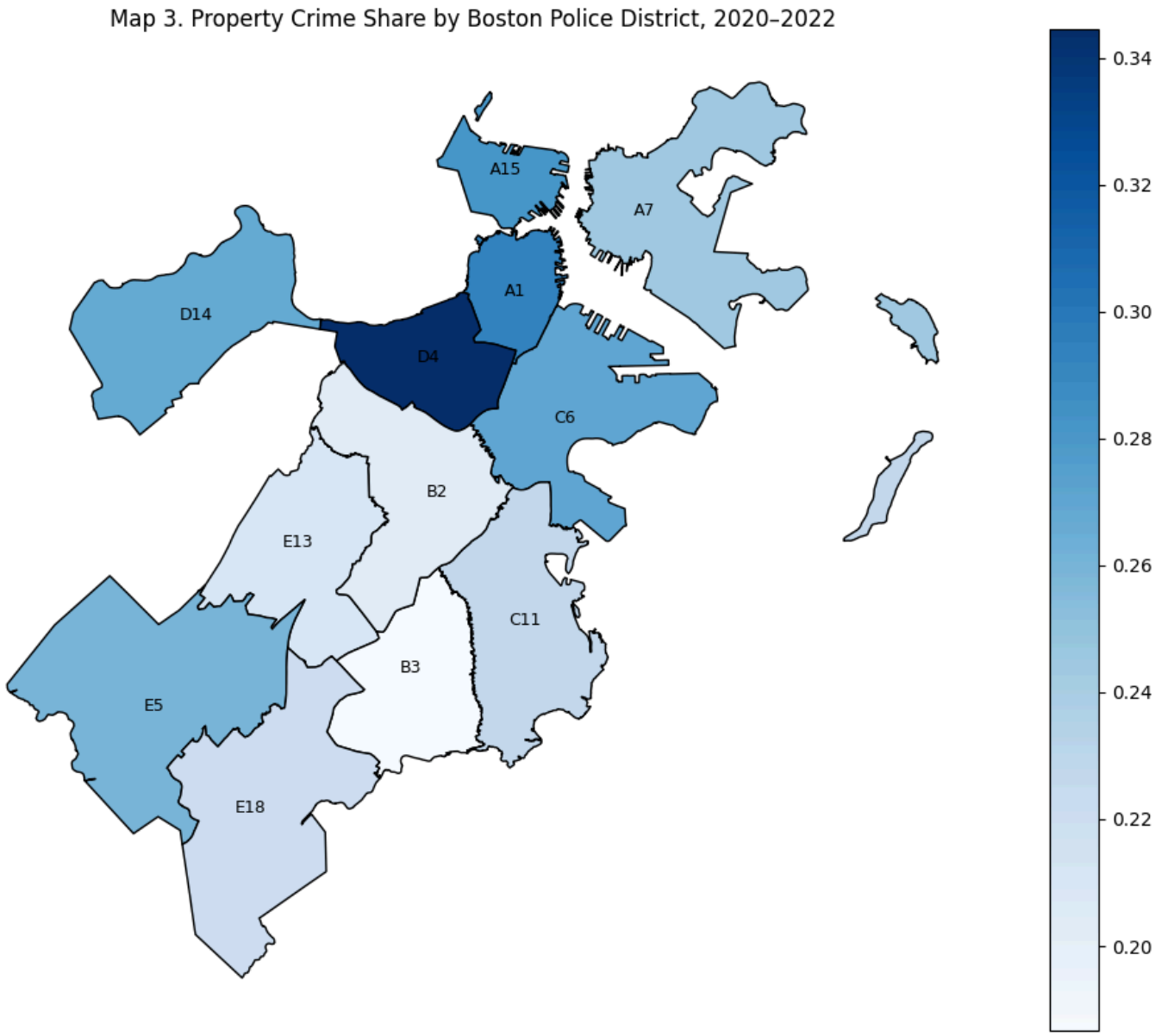
```
map3_df = crime_raw.copy()

map3_df["occurred_on_date"] = pd.to_datetime(map3_df["occurred_on_date"], errors="coerce")
map3_df = map3_df.dropna(subset=["occurred_on_date", "district"])
map3_df["year"] = map3_df["occurred_on_date"].dt.year
map3_df["district"] = map3_df["district"].astype(str).str.strip()

map3_df = map3_df[(map3_df["year"] >= 2020) & (map3_df["year"] <= 2022)]
```

```python
map3_df["offense_description"] = (
    map3_df["offense_description"]
    .astype(str)
    .str.lower()
    .str.strip()
)

map3_df["crime_group"] = map3_df["offense_description"].apply(crime_group)

district_totals = (
    map3_df.groupby("district")
    .size()
    .reset_index(name="total_incidents")
)

property_counts = (
    map3_df[map3_df["crime_group"] == "property_related"]
    .groupby("district")
    .size()
    .reset_index(name="property_incidents")
)

map3_data = district_totals.merge(property_counts, on="district", how="left")
map3_data["property_incidents"] = map3_data["property_incidents"].fillna(0)
map3_data["property_share"] = map3_data["property_incidents"] / map3_data["total_incidents"]

district_map = gpd.read_file("boston_police_districts.geojson")

district_map["DISTRICT"] = district_map["DISTRICT"].astype(str).str.strip()

map3_geo = district_map.merge(
    map3_data,
    left_on="DISTRICT",
    right_on="district",
    how="left"
)
```

In [196…

```python
fig, ax = plt.subplots(figsize=(10, 8))

map3_geo.plot(
    column="property_share",
    cmap="Blues",
    linewidth=1,
    edgecolor="black",
    legend=True,
    ax=ax,
    missing_kwds={"color": "lightgrey", "label": "Missing"}
)

for idx, row in map3_geo.iterrows():
    if row.geometry is not None:
        centroid = row.geometry.representative_point()
        ax.annotate(
            text=row["DISTRICT"],
            xy=(centroid.x, centroid.y),
            ha="center",
            fontsize=9
        )

ax.set_title("Map 3. Property Crime Share by Boston Police District, 2020–2022")
ax.axis("off")
```

```
plt.tight_layout()
plt.show()
```



Map 3 shows the share of reported incidents classified as property crime across Boston police districts during the 2020–2022 period. This map is especially important because it connects directly to the composition focus of the paper. The darker districts indicate places where property crime made up a larger share of total incidents during the COVID and post-COVID years. For example, districts such as D4 appear to have a relatively high property-crime share compared with several outer districts. This suggests that the pandemic period changed not only total crime levels, but also the mix of reported incidents across neighborhoods. In that sense, the map supports the broader argument that COVID reshaped the composition of crime unevenly across the city.

2.3 Regressions

In [197…

```
import statsmodels.formula.api as smf
from stargazer.stargazer import Stargazer
from IPython.display import HTML, display

reg_df = merged_df.copy()

# Full sample for baseline models
base_df = reg_df.dropna(subset=[
```

```
    "total_crime",
    "unemployment_rate",
    "covid",
    "trend",
    "labor_force"
]).copy()

# Shorter mobility sample
mob_df = reg_df.dropna(subset=[
    "total_crime",
    "unemployment_rate",
    "covid",
    "trend",
    "labor_force",
    "transit_stations_percent_change_from_baseline"
]).copy()
```

In [198…

```
m1 = smf.ols(
    "total_crime ~ unemployment_rate + covid + trend",
    data=base_df
).fit()

m2 = smf.ols(
    "total_crime ~ unemployment_rate + covid + labor_force",
    data=base_df
).fit()

m3 = smf.ols(
    "total_crime ~ unemployment_rate + covid + trend + labor_force",
    data=base_df
).fit()

m4 = smf.ols(
    "total_crime ~ unemployment_rate + covid + trend + I(unemployment_rate**2)",
    data=base_df
).fit()

stargazer1 = Stargazer([m1, m2, m3, m4])
stargazer1.title("Table 2. Baseline OLS Specifications")
stargazer1.rename_covariates({
    "Intercept": "Constant",
    "unemployment_rate": "Unemployment rate (%)",
    "covid": "COVID period",
    "trend": "Time trend",
    "labor_force": "Labor force",
    "I(unemployment_rate ** 2)": "Unemployment rate squared"
})

display(HTML(stargazer1.render_html()))
```

Table 2. Baseline OLS Specifications

				Dependent variable: total_crime
	(1)	(2)	(3)	(4)
Unemployment rate squared				-7.472
				(10.972)
Constant	8401.891***	-3613.575	-5375.280	7974.354***
	(331.718)	(8713.483)	(8373.514)	(710.793)
COVID period	-310.977	-1283.284***	28.691	-613.701
	(561.141)	(266.637)	(589.998)	(717.998)
Labor force		0.004	0.005	
		(0.003)	(0.003)	
Time trend	-31.887**		-34.624**	-26.263
	(14.150)		(14.037)	(16.442)
Unemployment rate (%)	-117.864**	-27.115	-99.892**	24.551
	(46.671)	(38.547)	(47.252)	(214.315)
Observations	60	60	60	60
R^2	0.594	0.570	0.613	0.597
Adjusted R^2	0.572	0.547	0.585	0.568
Residual Std. Error	718.716 (df=56)	739.419 (df=56)	707.980 (df=55)	722.182 (df=55)
F Statistic	27.282*** (df=3; 56)	24.745*** (df=3; 56)	21.765*** (df=4; 55)	20.381*** (df=4; 55)
Note:				*p<0.1; **p<0.05; ***p<0.01

Table 2 reports the baseline specifications using the full 60-month sample. Across these models, the clearest result is the negative time trend. In Models 1 and 3, the trend coefficient is about -31.9 and -34.6, both statistically significant, suggesting that reported crime was already declining by roughly 32 to 35 incidents per month on average over the sample period. The unemployment rate is also negative and significant in Models 1 and 3, with coefficients of about -117.9 and -99.9. This means higher unemployment is associated with lower reported crime in these specifications, which does not fit a simple economic-distress story.

The COVID coefficient is less stable across the baseline models. In Model 2, where COVID is included with unemployment and labor force but without a time trend, the coefficient is large, negative, and statistically significant at about -1,283. Once the time trend is added, the COVID coefficient becomes much smaller and loses significance, suggesting that part of the COVID effect is absorbed by the broader downward trend in the data. Labor force is positive in the baseline models but not statistically significant, so its role appears weaker than that of trend and unemployment.

My preferred specification is Model 3, which includes unemployment rate, the COVID dummy, a time trend, and labor force. This specification has the highest adjusted R-squared among the baseline models and includes all theoretically motivated variables while avoiding the redundancy introduced by the squared unemployment term in Model 4.

In [263…

```
m5 = smf.ols(
    "total_crime ~ unemployment_rate + covid + " \
    "transit_stations_percent_change_from_baseline",
    data=mob_df
).fit()

m6 = smf.ols(
    "total_crime ~ unemployment_rate + covid +" \
    " trend + transit_stations_percent_change_from_baseline",
    data=mob_df
).fit()

m7 = smf.ols(
    "total_crime ~ unemployment_rate + covid +" \
    " labor_force + transit_stations_percent_change_from_baseline",
    data=mob_df
).fit()

m8 = smf.ols(
    "total_crime ~ unemployment_rate + covid +" \
    " trend + labor_force + transit_stations_percent_change_from_baseline",
    data=mob_df
).fit()

stargazer2 = Stargazer([m5, m6, m7, m8])
stargazer2.title("Table 3. Mobility and Expanded OLS Specifications")
stargazer2.rename_covariates({
    "Intercept": "Constant",
    "unemployment_rate": "Unemployment rate (%)",
    "covid": "COVID period",
    "trend": "Time trend",
    "labor_force": "Labor force",
    "transit_stations_percent_change_from_baseline": "Transit mobility change (%)"
})

display(HTML(stargazer2.render_html()))
```

Table 3. Mobility and Expanded OLS Specifications

	Dependent variable: total_crime			
	(1)	(2)	(3)	(4)
Constant	5817.340***	-1243.746	-22853.192*	-4801.054
	(775.876)	(1797.195)	(9399.557)	(14419.445)
COVID period	2187.323	4437.828**	-722.424	3708.593
	(2279.014)	(1390.209)	(1810.639)	(3296.581)
Labor force			0.010**	0.002
			(0.003)	(0.006)
Transit mobility change (%)	38.257	128.192***	-25.884	107.113
	(46.776)	(34.314)	(37.944)	(92.530)
Time trend		243.031***		212.872
		(60.027)		(137.624)
Unemployment rate (%)	24.753	296.410**	-20.128	255.758
	(123.311)	(96.206)	(84.603)	(193.967)
Observations	11	11	11	11
R^2	0.215	0.790	0.693	0.792
Adjusted R^2	-0.122	0.649	0.488	0.584
Residual Std. Error	698.745 (df=7)	390.683 (df=6)	472.126 (df=6)	425.342 (df=5)
F Statistic	0.637 (df=3; 7)	5.627** (df=4; 6)	3.380* (df=4; 6)	3.810* (df=5; 5)
Note:				*p<0.1; **p<0.05; ***p<0.01

Table 3 adds the mobility variable and speaks more directly to the revised focus of the project. Because these models use only 11 observations, they should be read as suggestive rather than conclusive. Even so, they offer useful evidence. In Model 2, transit mobility has a positive and statistically significant coefficient of about 128.2, suggesting that months with higher transit mobility are associated with higher reported crime. In the same model, the time trend and unemployment rate are also positive and significant. This supports the paper's main message: in the short run, crime appears to move more closely with changes in mobility and routine activity than with unemployment alone.

The mobility results are not fully stable across all four models. The mobility coefficient is not statistically significant in Models 1, 3, or 4, and the COVID coefficient also changes considerably across specifications. These shifts are likely due in part to the short mobility sample, which makes the expanded regressions more sensitive to specification changes. Still, the directional evidence from Model 2 is consistent with the descriptive figures and supports the paper's main argument.

Project 3

3.1 Potential Data to Scrape

To add information that is not already in my crime dataset, I scrape COVID-19 data from the Wikipedia page on the COVID-19 pandemic in Massachusetts. The webpage address is:

https://en.wikipedia.org/wiki/COVID-19_pandemic_in_Massachusetts

This source ties to my research question because it helps identify when the pandemic shock was most intense and compare those periods to changes in crime and mobility.

The website contains multiple HTML tables. After reviewing them, I chose the table that reports dated COVID-related observations because it provides a clear time dimension. I scrape the date field and use the table structure to construct a monthly measure of scraped COVID activity. This does not replace my main crime regressions, but it provides additional time-based evidence that helps interpret the timing of the crime changes shown earlier in the paper.

I do not use this scraped dataset as a direct regression input in my preferred specification. Instead, I use it as supporting evidence to compare the timing of the broader COVID shock with movements in crime and mobility. And this is helpful for answering the research question because it shows whether the decline in crime occurred during the most disruptive periods of the pandemic.

3.2 Potential Challenges

One main challenge here is that the webpage is not designed as a clean research dataset. Although the page contains useful HTML tables, the columns are not immediately ready for analysis and some tables include duplicated or unhelpful fields. This means the scraped data must be cleaned before it can be used.

Another challenge is that the scraped source is better suited for descriptive analysis than for direct merging into the main regression framework. The table captures COVID-related reporting activity rather than a perfect measure of restrictions or mobility. For that reason, I use the scraped data mainly as contextual evidence to support the interpretation of my results.

I do not need to run the scraper repeatedly over time for this project because the analysis is historical and the relevant COVID period is already covered. However, if the paper were extended in the future, similar scraping could be done repeatedly on policy or news pages to build a richer event-based dataset.

3.3 Scraping Data from a Website

The first step is to access the webpage and extract its HTML tables. I use the `requests` package to retrieve the page and confirm that the request succeeds. Then I use `pandas.read_html()` to read the HTML tables from the page.

After identifying the table that contains dated COVID-related information, I clean the column names and keep only the date field. Since the table was not perfectly clean for direct use as a case-count dataset, I instead transform it into a monthly measure of scraped COVID activity by counting the number of dated table entries in each month.

This approach still produces a useful time series from the scraped webpage. It allows me to compare when COVID-related reporting activity was most intense relative to the movements in crime shown in my main dataset.

In [200…

```
import pandas as pd
import requests
from io import StringIO

url = "https://en.wikipedia.org/wiki/COVID-19_pandemic_in_Massachusetts"
headers = {"User-Agent": "Mozilla/5.0"}

response = requests.get(url, headers=headers, timeout=20)
print("Status:", response.status_code)

html_data = StringIO(response.text)
tables = pd.read_html(html_data)

print("Number of tables:", len(tables))
for i, table in enumerate(tables):
    print(f"\nTABLE {i}")
    print("Shape:", table.shape)
    print("Columns:", list(table.columns))
```

```
Status: 200
Number of tables: 15

TABLE 0
Shape: (14, 2)
Columns: ['COVID-19 pandemic in Massachusetts', 'COVID-19 pandemic in Massachusetts.1']

TABLE 1
Shape: (1109, 5)
Columns: ['Date', 'Date.1', 'Unnamed: 2', '# of cases', '# of deaths']

TABLE 2
Shape: (15, 7)
Columns: [('County [a]', '14 / 14'), ('Confirmed Cases [b]', '1,305,830'), ('Deaths [c]', '20,87
2'), ('Population[d]', '6,892,503'), ('Cases /100k', '18,945.7'), ('Deaths /100k', '302.8'), ('De
aths /Case\xa0%', '1.6')]

TABLE 3
Shape: (1, 2)
Columns: [0, 1]

TABLE 4
Shape: (1, 2)
Columns: [0, 1]

TABLE 5
Shape: (1, 2)
Columns: [0, 1]

TABLE 6
Shape: (1, 2)
Columns: [0, 1]

TABLE 7
Shape: (22, 2)
Columns: ['vteCOVID-19 pandemic in the United States', 'vteCOVID-19 pandemic in the United State
s.1']

TABLE 8
Shape: (3, 2)
Columns: [0, 1]

TABLE 9
Shape: (6, 2)
Columns: [0, 1]

TABLE 10
Shape: (2, 2)
Columns: [0, 1]

TABLE 11
Shape: (2, 2)
Columns: [0, 1]

TABLE 12
Shape: (4, 2)
Columns: [0, 1]

TABLE 13
Shape: (2, 2)
Columns: [0, 1]
```

```
TABLE 14
Shape: (3, 2)
Columns: [0, 1]
```

After extracting the HTML tables, I inspect the available tables and select the one with the clearest time structure. Among the available tables, the most useful one is the table containing dated COVID-related entries. This table is the most suitable for comparing the timing of the pandemic shock to the crime timeline in my project.

In [201…

```
covid_df = tables[1].copy()
covid_df.columns = ["date", "date_dup", "junk", "cases", "deaths"]

covid_df["date"] = pd.to_datetime(covid_df["date"], errors="coerce")
covid_df = covid_df.dropna(subset=["date"]).copy()

covid_df["year"] = covid_df["date"].dt.year
covid_df["month"] = covid_df["date"].dt.month
covid_df["covid_update"] = 1

covid_monthly = (
    covid_df.groupby(["year", "month"])["covid_update"]
    .sum()
    .reset_index()
)

covid_monthly["date"] = pd.to_datetime(
    dict(year=covid_monthly["year"], month=covid_monthly["month"], day=1)
)

covid_monthly = covid_monthly[
    (covid_monthly["date"] >= "2018-01-01") &
    (covid_monthly["date"] <= "2022-12-01")
].copy()

covid_monthly.head()
```

Out[201…

	year	month	covid_update	date
0	2020	2	1	2020-02-01
1	2020	3	28	2020-03-01
2	2020	4	30	2020-04-01
3	2020	5	31	2020-05-01
4	2020	6	30	2020-06-01

The cleaned output above shows the scraped data aggregated to the monthly level. The variable `covid_update` counts the number of dated scraped entries in each month. I use this as a simple measure of COVID-related reporting activity over time.

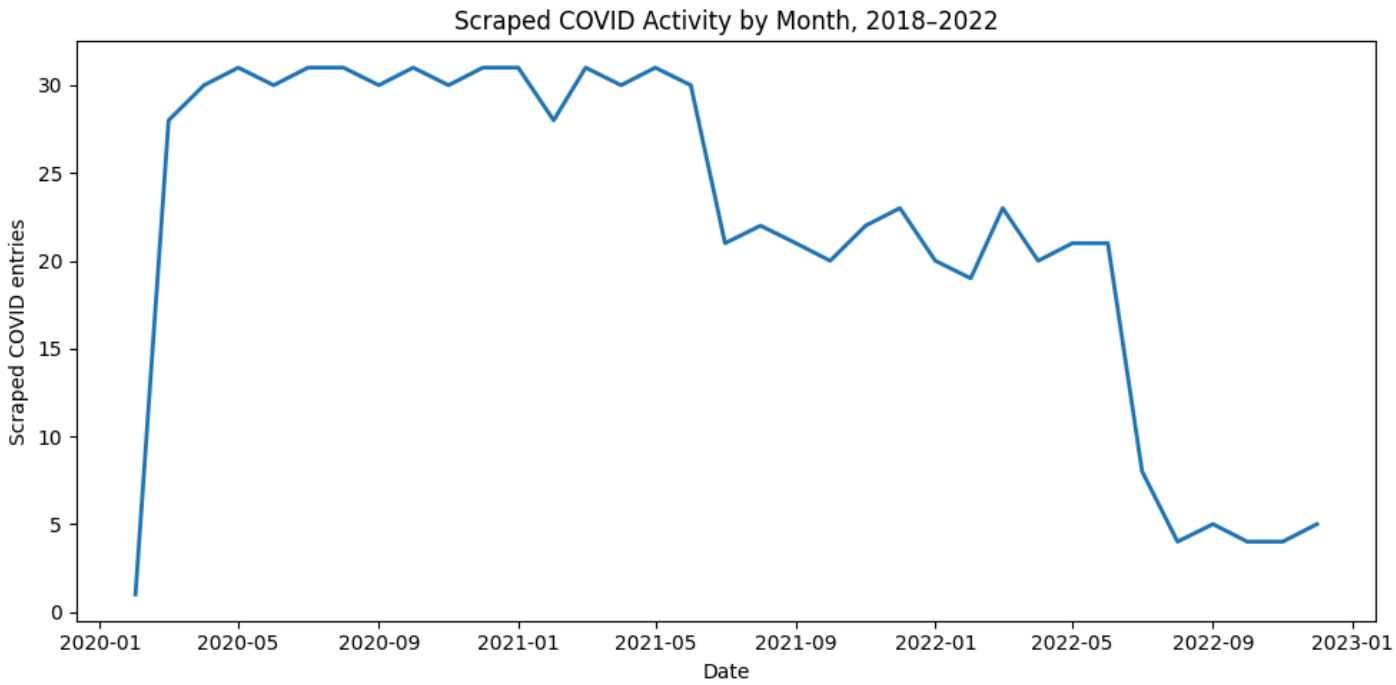
3.4 Visualizing the Scraped Dataset

I use the scraped data to create a monthly time series of COVID-related activity and compare it to reported crime in Boston. The goal is not to treat the scraped variable as a perfect measure of pandemic severity, but rather to use it as additional time-based evidence about when the COVID shock was most active.

The first visualization shows scraped COVID activity by month. The second compares that scraped activity to monthly reported crime in Boston. Together, these figures help show that the sharpest crime changes occurred during periods of intense pandemic disruption.

In [203…

```
import matplotlib.pyplot as plt

plt.figure(figsize=(10,5))
plt.plot(covid_monthly["date"], covid_monthly["covid_update"], linewidth=2)
plt.title("Scraped COVID Activity by Month, 2018–2022")
plt.xlabel("Date")
plt.ylabel("Scraped COVID entries")
plt.tight_layout()
plt.show()
```

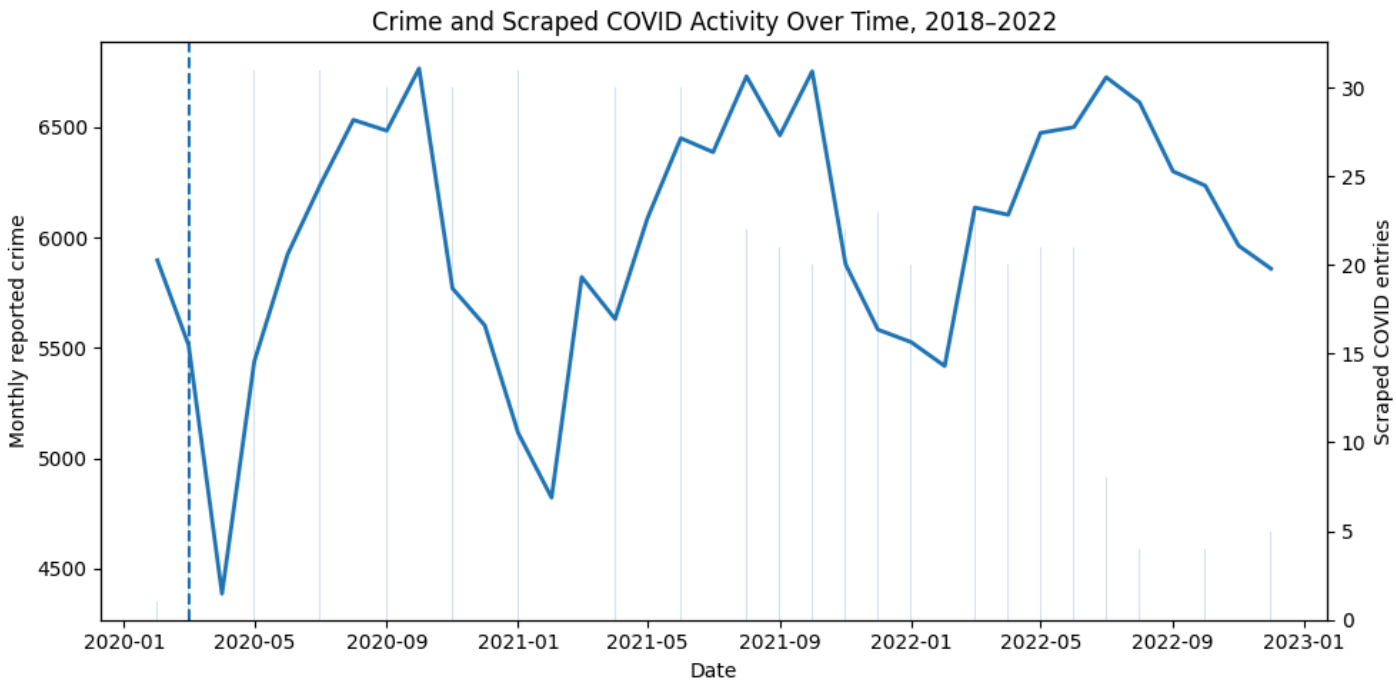


This figure shows that scraped COVID-related activity is concentrated during the core pandemic period, especially in 2020 and 2021. This is useful because it identifies when the broader COVID shock was most active, which can then be compared to the timing of changes in crime and mobility.

In [204…

```
fig, ax1 = plt.subplots(figsize=(10,5))

ax1.plot(
    scrape_compare["date"],
    scrape_compare["total_crime"],
    linewidth=2,
    label="Total crime"
)
ax1.set_ylabel("Monthly reported crime")
ax1.set_xlabel("Date")
ax1.axvline(pd.to_datetime("2020-03-01"), linestyle="--", linewidth=1.5)

ax2 = ax1.twinx()
ax2.bar(
    scrape_compare["date"],
    scrape_compare["covid_update"],
    alpha=0.2
)
ax2.set_ylabel("Scraped COVID entries")
```

```
plt.title("Crime and Scraped COVID Activity Over Time, 2018–2022")
plt.tight_layout()
plt.show()
```



This figure compares monthly reported crime in Boston to the number of scraped COVID-related table entries between 2020 and 2022. The scraped series is highest during 2020 and remains elevated through much of 2021, which is the same broad period in which crime falls sharply relative to pre-pandemic levels. While the scraped variable is not a direct measure of restrictions and is not used in the main regression, it provides supporting time-based evidence that the largest changes in crime occurred during the most disruptive phase of the COVID shock. This reinforces the paper's main interpretation that disruptions to mobility and routine activity played an important role in shaping crime patterns.

3.5 Machine Learning

3.5.1 Regression Tree

To explore nonlinear relationships between crime and its predictors, I estimate a regression tree. A regression tree partitions the data into regions such that variation in the outcome variable within each region is minimized. Formally, the objective is:

$$\min_{\text{splits}} \sum_{m=1}^{M} \sum_{i \in R_m} (y_i - \bar{y}_{R_m})^2$$

where R_m represents a leaf of the tree and $\bar{y}_{R_m}$ is the mean of monthly crime within that leaf.

In simple terms, the tree searches for cutoff values in the explanatory variables that split the data into groups with similar crime levels. This is useful because the relationship between crime and its predictors may not be fully linear during the COVID period.

3.5.2 Regularization Parameters

Regression trees can overfit the data if they are allowed to grow too much. To reduce that risk, I use regularization parameters.

- **max_depth** limits how deep the tree can grow
- **min_samples_split** sets the minimum number of observations needed for a split
- **min_samples_leaf** sets the minimum number of observations in each terminal leaf

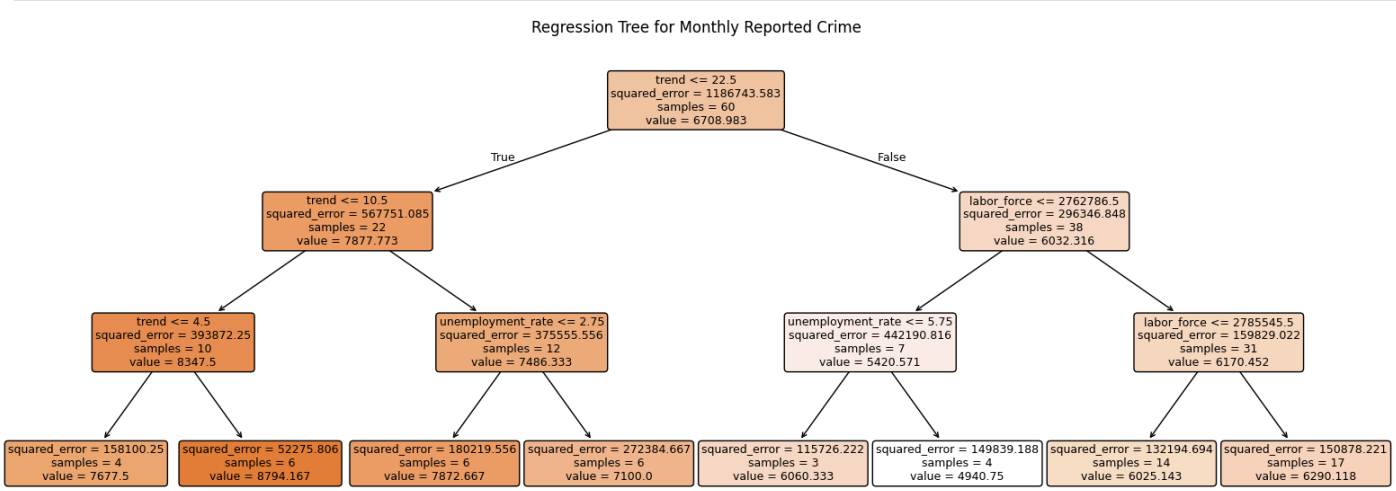
Increasing these values makes the model simpler and more stable, while decreasing them allows the tree to fit more complex patterns but increases the risk of overfitting. In this project, I keep the tree relatively shallow so that it remains interpretable and does not fit noise in the data.

3.5.3 Preferred Specification Tree

I first estimate a regression tree using the main explanatory variables from my preferred regression specification. These include unemployment rate, labor force, the COVID dummy, and a time trend. This keeps the machine learning section consistent with the main regression framework and makes the comparison with OLS more meaningful.

In [243…

```
df_tree = merged_df[features_pref + ["total_crime"]].dropna().copy()

X_pref = df_tree[features_pref]
y_pref = df_tree["total_crime"]

tree_pref = DecisionTreeRegressor(
    max_depth=3,
    random_state=42
)

tree_pref.fit(X_pref, y_pref)

plt.figure(figsize=(20, 7))
plot_tree(
    tree_pref,
    feature_names=features_pref,
    filled=True,
    rounded=True,
    fontsize=9
)
plt.title("Regression Tree for Monthly Reported Crime")
plt.show()
```



The preferred regression tree reveals threshold effects in the data. The earliest splits are driven mainly by the time trend and labor force conditions, which suggests that crime responds differently across different stages of the sample and under different labor market conditions. This supports the broader argument of the paper that crime during the pandemic period cannot be fully described by one single linear relationship.

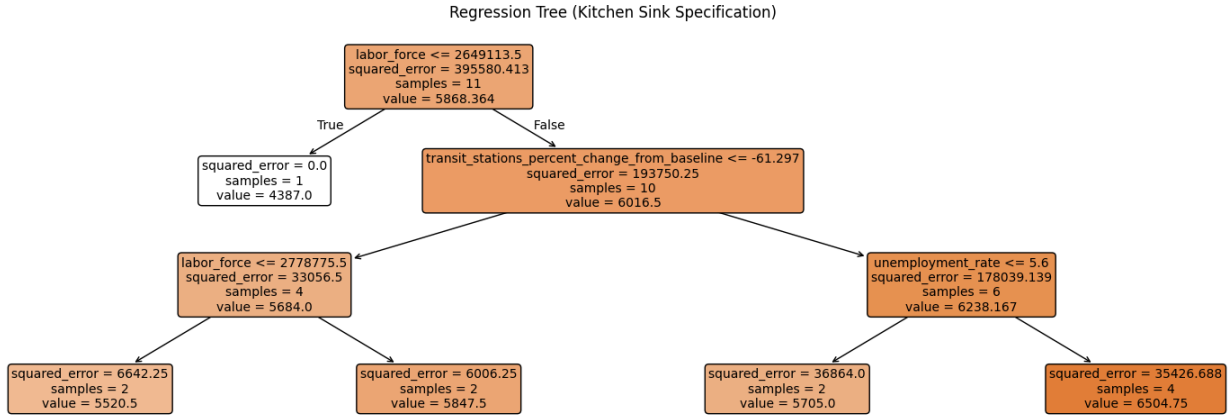
Unlike OLS, the tree allows the relationship between crime and its predictors to vary across regions of the data. In that sense, it adds useful descriptive evidence about nonlinear patterns, although it should not be interpreted as a causal model.

3.5.4 Kitchen Sink Tree

I next estimate a broader tree using all available explanatory variables, including transit mobility. The goal is to see whether the main split variables change once the model is given a wider set of predictors.

In [248…

```
features_all = [
    "unemployment_rate",
    "labor_force",
    "covid",
    "trend",
    "transit_stations_percent_change_from_baseline"
]

df_tree_all = merged_df[features_all + ["total_crime"]].dropna().copy()

X_all = df_tree_all[features_all]
y_all = df_tree_all["total_crime"]

tree_all = DecisionTreeRegressor(
    max_depth=3,
    random_state=42
)

tree_all.fit(X_all, y_all)

plt.figure(figsize=(20, 6))
plot_tree(
    tree_all,
    feature_names=features_all,
    filled=True,
    rounded=True,
    fontsize=10
)
plt.title("Regression Tree (Kitchen Sink Specification)")
plt.show()
```



Including all variables changes the structure of the tree somewhat, but the main split variables remain time-related and labor-market variables. The mobility variable appears in the broader specification, but because mobility is only available for a short 2020 window, this kitchen sink tree is estimated on a much smaller sample and should be interpreted cautiously.

Overall, the comparison suggests that the main descriptive patterns are fairly stable: crime varies importantly with time and labor market conditions, and the relationship is not fully linear.

3.5.5 Error of Prediction

To evaluate how well the preferred tree fits the data, I compute the mean squared error of prediction.

In [249…

```
from sklearn.metrics import mean_squared_error

y_pred_tree = tree_pref.predict(X_pref)
mse_tree = mean_squared_error(y_pref, y_pred_tree)

print("Tree MSE:", mse_tree)
```

```
Tree MSE: 150397.86770541547
```

The prediction error shows that the tree captures meaningful structure in the data, but it does not fully explain monthly crime on its own. This is expected, since crime is affected by many factors not included in the model. In smaller samples, tree-based models can also become sensitive to particular splits, which is another reason to interpret the results as descriptive rather than definitive.

3.6 Random Forest

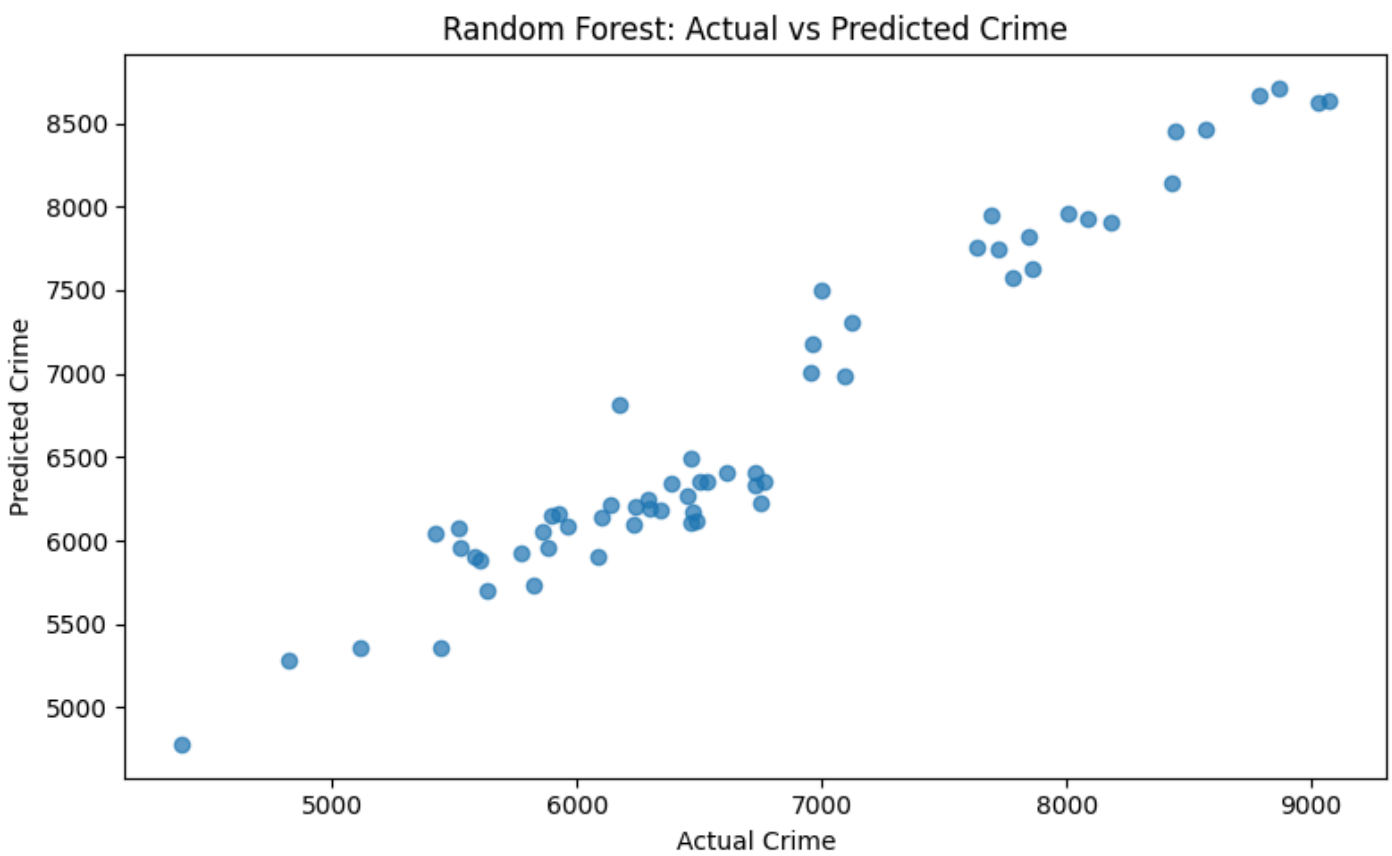
3.6.1 Model Setup and Prediction

I next estimate a random forest model. A random forest improves on a single tree by averaging across many trees, which reduces instability and usually improves predictive performance. I estimate the main random forest on the broader full sample using unemployment rate, labor force, the COVID dummy, and the time trend.

In [250…

```
from sklearn.ensemble import RandomForestRegressor
import pandas as pd
import matplotlib.pyplot as plt

rf_df = merged_df.copy()
```

```python
features_rf = [
    "unemployment_rate",
    "labor_force",
    "covid",
    "trend"
]

target = "total_crime"

rf_df = rf_df[features_rf + [target]].dropna().copy()

X_rf = rf_df[features_rf]
y_rf = rf_df[target]

rf_model = RandomForestRegressor(
    n_estimators=200,
    max_depth=4,
    random_state=42
)

rf_model.fit(X_rf, y_rf)

y_pred_rf = rf_model.predict(X_rf)

plt.figure(figsize=(8, 5))
plt.scatter(y_rf, y_pred_rf, alpha=0.7)
plt.xlabel("Actual Crime")
plt.ylabel("Predicted Crime")
plt.title("Random Forest: Actual vs Predicted Crime")
plt.tight_layout()
plt.show()
```



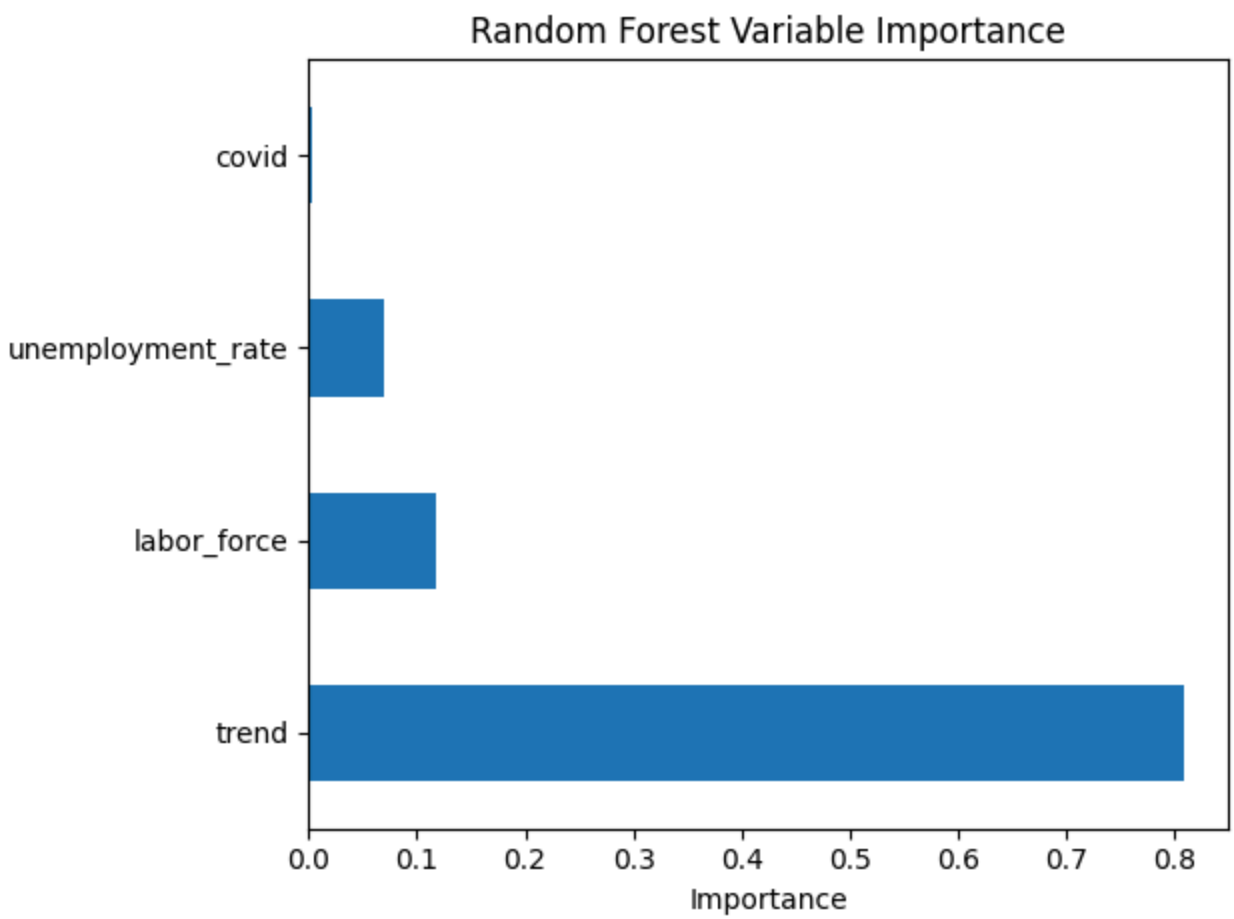
The actual-versus-predicted figure shows that the random forest captures broad variation in monthly crime reasonably well. The model does not produce perfect predictions, but it improves stability relative to a single

tree by combining information across many trees.

In [253…

```
importance = pd.Series(rf_model.feature_importances_, index=features_rf)
importance = importance.sort_values(ascending=False)

print(importance)

importance.plot(kind="barh")
plt.title("Random Forest Variable Importance")
plt.xlabel("Importance")
plt.tight_layout()
plt.show()
```

```
trend                0.809104
labor_force          0.118012
unemployment_rate    0.069220
covid                0.003664
dtype: float64
```



The random forest importance matrix shows that the **time trend** is by far the most important predictor, followed by **labor force** and **unemployment rate**. The **COVID dummy** has near-zero importance once the other variables are included.

3.6.3 Interpretation

These results suggest that crime during this period is explained more by broader time-related structure and labor-market conditions than by the COVID dummy alone. In other words, the pandemic appears to matter less as a simple binary event and more through the broader changes that occurred over time.

This interpretation is consistent with the earlier regression results. The OLS models already suggested that crime was trending downward and that unemployment did not operate through a simple distress

mechanism. The random forest adds to that story by showing that the overall time structure of the period is the strongest predictor in the data.

3.7 OLS vs ML

The OLS regressions and machine learning models provide complementary insights.

OLS is useful because it gives interpretable coefficients and shows average relationships between crime and the explanatory variables. In the baseline regressions, crime trends downward over time and higher unemployment is not associated with higher crime during this period.

The machine learning models add information that OLS cannot capture as easily. The regression tree reveals threshold effects, while the random forest ranks variables by predictive importance. In my results, the time trend is the dominant predictor and the COVID dummy contributes very little.

Taken together, the machine learning results reinforce the main conclusion from the regression analysis: crime during the COVID period appears to be shaped more by broad time-related and activity-related changes than by a simple economic distress mechanism alone.

3.8 Conclusion

This project examined how reported crime in Boston changed during the COVID-19 period, with a particular focus on whether the pandemic altered not only the level of crime but also its composition. The main finding is that reported crime fell sharply during the initial lockdown period, even as unemployment rose. This pattern is difficult to explain using a simple economic-distress framework alone. Instead, the evidence suggests that the pandemic changed crime partly by disrupting routine activity and reducing movement through the city.

The main message figure captures this clearly. Around March 2020, crime drops at the same time that transit mobility collapses and unemployment increases. Read together, these series suggest that the short-run effect of lockdowns and reduced movement may have been stronger than the effect of worsening labor market conditions. Fewer people moving through public space likely meant fewer opportunities for some types of crime, even during a period of substantial economic stress.

The spatial evidence supports this interpretation. The maps show that crime was not evenly distributed across Boston police districts and that the pandemic did not affect all areas in the same way. Some districts experienced much sharper declines in reported crime than others, while the property crime share map shows that the pandemic period was associated not only with lower overall crime in some places but also with changes in the mix of incidents recorded across districts.

The regression results point in the same broad direction. In the baseline models, crime trends downward over time and higher unemployment is not associated with higher reported crime during this period. In the mobility models, transit mobility is positively associated with crime in one of the stronger specifications, supporting the idea that movement through the city mattered for short-run crime outcomes. Because the mobility sample only covers part of 2020, these results should be read as suggestive rather than conclusive.

The machine learning results add further texture. The regression tree identifies nonlinear threshold effects, while the random forest shows that the time trend is the most important predictor, followed by labor force and unemployment. The COVID dummy alone has near-zero importance. Together, these results suggest

that the pandemic mattered less as a simple binary event and more through broader structural changes over time.

Overall, the project suggests that COVID changed crime in Boston through more than one channel. The pandemic was both an economic shock and a mobility shock, but the evidence here indicates that disruptions to routine activity and urban movement may have been more important for short-run crime patterns than unemployment alone.

There are important limitations to acknowledge. The mobility analysis is based on a short sample, since the Google Mobility dataset available for this project only covers part of 2020. In future work, a longer mobility series would allow for a more convincing test of the mechanism across the full sample period. It would also be useful to build offense-specific models and collect additional scraped data on policy intensity or restriction stringency to better identify the timing and magnitude of the COVID shock.

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